Sri Chaitanya ACADEMY | Infinity Learn



JEE FINAL TEST SERIES

JEE Advanced Full Syllabus Test - 4 [2023 P1]

Test Date: 16th Apr 2025

INSTRUCTIONS

- Use the OMR sheet provided separately for answering the questions.
- Blank spaces are provided within this booklet for rough work.
- Write your roll number in the space provided on the front cover of this Question Paper
- You are allowed to take away the Question Paper at the end of the examination.
- The OMR will be collected by the invigilator at the end of the examination.
- Do not tamper with or mutilate the OMR. Do not use the OMR for rough work.
- Write your name, roll number and sign with pen in the space provided for this purpose on the OMR.
- Do not write any of these details anywhere else on the OMR. Darken the appropriate bubble under each digit of your roll number.
- Use a BLACK BALL POINT PEN to darken the bubbles on the OMR>
- The OMR is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- Darken the bubbles ONLY IF you are sure of the answer. There is NO WAY to erase or "un-darken" a darkened bubble.

JEE-ADVANCED-2023-P1-Model

Time: 3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No. of Qs	Total marks
Sec – I(Q.N : 1 – 3)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – II(Q.N : 4 – 7)	Questions with Single Correct Choice	+3	-1	4	12
Sec – III(Q.N : 8 – 13)	Questions with **NON-NEGATIVE INTEGER** Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Matching Type	+3	-1	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No. of Qs	Total marks
Sec – I(Q.N : 18 – 20)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – II(Q.N : 21 – 24)	Questions with Single Correct Choice	+3	-1	4	12
Sec – III(Q.N : 25 – 30)	Questions with **NON-NEGATIVE INTEGER** Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Matching Type	+3	-1	4	12
Total				**17**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No. of Qs	Total marks
Sec – I(Q.N : 35 – 37)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – II(Q.N : 38 – 41)	Questions with Single Correct Choice	+3	-1	4	12
Sec – III(Q.N : 42 – 47)	Questions with **NON-NEGATIVE INTEGER** Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Matching Type	+3	-1	4	12
Total				**17**	**60**

MATHEMATICS

Max Marks: 60

SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

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1. If a, b, c are positive real numbers and the minimum value of

$$\frac{(a+3c)}{(a+c+2b)}+\frac{4b}{(a+b+2c)}-\frac{8c}{(a+b+3c)} \text{ is } k\sqrt{2}-17 \text{ then k is more than or equal to:}$$

A) 8 **B)** 10 **C)** 12 **D)** 14

2. If two tangents are drawn from origin upon the curve $f(x)=(\ell nx)^2+2\ell nx$. Then which of the following is/are **TRUE**?

A) Product of abscissae of points of contacts is 1.

B) Product of abscissae of points of contacts is e^2.

C) Number of solution of equation $x\left(5f'(x)-\ell n10\right)=10$ is 1.

D) Number of solutions of equation $x\left(5f'(x)-\ell n10\right)=10$ are 2.

3. If given $\vec{a}=\vec{b}+\vec{c}$. There is a non–zero vector $\vec{d}$ which is parallel to $\vec{b}$ and perpendicular to $\vec{c}$. Also given $|\vec{c}|=2$. Then which of the following is/are **TRUE**?

A) $\dfrac{\left|\vec{d}\times\left(\vec{a}\times\vec{d}\right)\right|}{\left|\vec{d}\right|^2}=1$ **B)** $\dfrac{\left|\vec{d}\times\left(\vec{a}\times\vec{d}\right)\right|}{\left|\vec{d}\right|^2}=2$ **C)** $\left|\dfrac{\vec{b}.\vec{a}}{\vec{b}.\vec{d}}\right|=\dfrac{|\vec{b}|}{|\vec{d}|}$ **D)** $\left|\dfrac{\vec{b}.\vec{a}}{\vec{b}.\vec{d}}\right|=\dfrac{2.|\vec{b}|}{|\vec{d}|}$

SECTION – II
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

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4. $f(x)=\sum_{r=1}^{n}\left(x^r+x^{-r}\right)^2$; $x\neq\pm1$; $g(x)=\begin{cases}\lim\limits_{n\to\infty}\left(\left(f(x)-2n\right)x^{-2n-2}.\left(1-x^2\right)\right) & x\neq\pm1\\ 1 & x=\pm1\end{cases}$

Then which of the following is **INCORRECT** for g (x)?

A) Discontinuity at x = –1 **B)** Continuity at x = 2
C) Removable discontinuity at x = 1 **D)** Continuity at x = –2

5. Let: $\int_0^{\pi}\frac{x.dx}{1-\cos\phi.\sin x}=f(\pi-\phi)$ then find value of $\int_0^{\frac{\pi}{2}}f^2(x).dx$

A) $\pi^3\ell n2$ **B)** $2\pi^3\ell n2$ **C)** $\frac{\pi^3}{2}\ell n2$ **D)** $3\pi^3\ell n2$

6. Suppose f is twice differentiable function satisfying $f(0)=-2, f'(0)=11; f''(0)=-8, f(2)=5; f'(2)=-3; f''(2)=7$ and also suppose that the function $g(x)=\frac{1}{x}\int_0^x f(t)\,dt$ has a critical point at x = 2. Then

Statement-I: critical point of g (x) at x = 2 is local maxima,

Statement-II: $\int_0^2 f(t)\,dt=10$

A) only statement-I is correct **B)** only statement-II is correct
C) both statements are correct **D)** both statements are incorrect

7. The solution of the differentiable equation $x^2dy-y^2dx+xy^2(x-y)dy=0$ is f (x, y) = 0, where f (1, 2) = 0. The curve f (x, y) = 0 and hyperbola (x – 2) (y + 2) + 4 = 0 intersects at (α,β) the value of $\alpha+\beta$ is_____

A) 1 **B)** 3 **C)** 5 **D)** 7

SECTION-III
(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

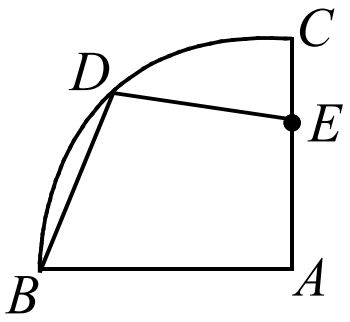
- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

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8. Variance of the data 1, 3, 5, x, y is two more than that of the data 1, 3, 5, x + 1, y – 1. Minimum value of x^2+y^2 is _____ $(x,y\in R)$

9. The given figure ABCA is a quater of a circle.



If BD = DE = 8 units and $\angle BDE = 90^\circ$, then length of CE is $a.\sqrt{b-\sqrt{c}}$. Find $a+b+c$. $(a,b,c \in N)$

10. The number of values of θ in the range $[0,2\pi]$ satisfying the equation $\cos^4 2\theta + 2\sin^2 2\theta = 17(\sin\theta + \cos\theta)^8$ is_____

11. The polynomial $f(z) = az^{4036} + bz^{4035} - cz^{4034}$ has real coefficients where a, b, c belongs to $(-\infty, 4038]$. If $f\left(e^{-\frac{i\pi}{3}}\right) = -4032 + 4038\sqrt{3}i$, then the value of $|a-b|$ is

12. Find the total number of real solutions of equation $e^{14x} - 2e^{7x} - e^{10x} - 6.e^{6x} + e^{3x} + 1 = 0$

13. For the given curve, $x = a\cos t + \frac{1}{2}a\,\ell n\,\tan^2\frac{t}{2}; y = a\sin t$. If the area bounded by the curve is $K\pi a^2$, then K is _____

SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

14. Match the following.

	Column–I		**Column–II**
A)	Q is image of P(2, 4) in bisector line L of 1st quadrant, S is circle with center Q and radius 4. A circle M is drawn such that it touch line L at (1, 1) and its radical axis with S passes through P then its radius is	**P)**	$12\sqrt{2}$
B)	Tangents are drawn from P(K, 0) on $x^2 + y^2 = 50$, meeting Y–axis at Q and R, then possible value of K for which area of ΔPQR is minimum, is:	**Q)**	$\sqrt{50}$

C)	If length of major axis of ellipse. $2x^2 + xy + y^2 = 5$ is 2K and $K^2 = \frac{10}{a-b}$, then $a^2.b$ is _____. (where a is an integer and b is irrational)	**R)**	$\frac{\sqrt{162}}{2}$
D)	Center of a circle passing through foci of $\frac{x^2}{25} - \frac{y^2}{24} = 1$ and through the points of intersection of tangents at vertices with tangent at point $\left(5\sqrt{2}, 2\sqrt{6}\right)$ is (h, k) then $\left\|2\sqrt{3}.k\right\|$ equals:	**S)**	10
		T)	$\sqrt{162}$

A) A–Q, B–R, C–P, D–T **B)** A–R, B–S, C–T, D–P

C) A–R, B–P, C–T, D–Q **D)** A–Q, B–S, C–S, D–P

15. If all unit squares of a chess board are painted red and blue at random then find probability that

Column–I		**Column–II**	
A)	Chess board contains equal number of red and blue squares.	**P)**	$\frac{1}{2^{63}}$
B)	All the squares in any column are of same colour and alternating colour in all rows.	**Q)**	$\frac{{}^{64}C_{32}}{2^{64}}$
C)	Not all squares in any column are alternating in colour.	**R)**	$\frac{1}{2^{15}}$
D)	Each row and each column contain odd number of blue squares.	**S)**	$\left(\frac{2^7-1}{2^7}\right)^8$
		T)	$\left(\frac{2^8-1}{2^8}\right)^8$

A) A–Q, B–P, C–S, D–R **B)** A–Q, B–P, C–T, D–P

C) A–R, B–P, C–S, D–P **D)** A–S, B–P, C–T, D–R

16. Match the following List–I with List–II (where $\left|X\right|$ denotes determinant of matrix X, X^T denotes transpose of matrix X, I denotes identity matrix)

List–I		**List–II**	
P)	$\left\|A_{4\times3}\ B_{3\times4}\right\|$ is equal to	**1)**	1
Q)	If A and B are non–zero square matrices of same order such that AB = O, then the value of $\left\|A\right\| + \left\|B\right\| + 5$ is equal to (where O is zero matrix)	**2)**	Not defined
R)	Let A be an 3×3 orthogonal matrix such that AB = BA, then $\left\|AB^T - B^T A + 2I\right\|$ is equal to	**3)**	0

S)	Let A and B be square matrices of order 3×3 such that $\lvert A\rvert=1$ and $\lvert B\rvert=3$, then the value of $\lvert 2A^2B^{-3}\rvert$ is equal to	**4)**	8
		5)	5
		6)	$\frac{8}{27}$

A) P–2, Q–6, R–4, S–1 **B)** P–2, Q–4, R–5, S–6

C) P–3, Q–5, R–2, S–6 **D)** P–3, Q–5, R–4, S–6

17. Let $\vec{a},\vec{b},\vec{c},\vec{d}$ are position vectors of 4 points A, B, C, D such that $\lvert\vec{a}\rvert=1,\lvert\vec{b}\rvert=2,\lvert\vec{c}\rvert=3,\lvert\vec{d}\rvert=4$ and $(\vec{a}\times\vec{b}).(\vec{c}\times\vec{d})=24,\vec{a}.\vec{c}=\frac{3}{2},\vec{b}.\vec{c}>0$, then area of $\Delta ABC=\frac{1}{4}\left(2\alpha+\beta\sqrt{3}\right)$ and area of $\Delta BCD=\frac{1}{2}\left(m-n\sqrt{3}\right)$, where $\alpha,\beta,m,n\in N$.

Match List–I with List–II and select the **CORRECT** answer using the code given below the list.

List–I		**List–II**	
P)	α	**1)**	1
Q)	β	**2)**	3
R)	m	**3)**	4
S)	n	**4)**	9

A) P–1, Q–2, R–3, S–4 **B)** P–2, Q–1, R–3, S–4

C) P–2, Q–1, R–4, S–3 **D)** P–1, Q–2, R–4, S–3

PHYSICS

Max Marks: 60

SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)

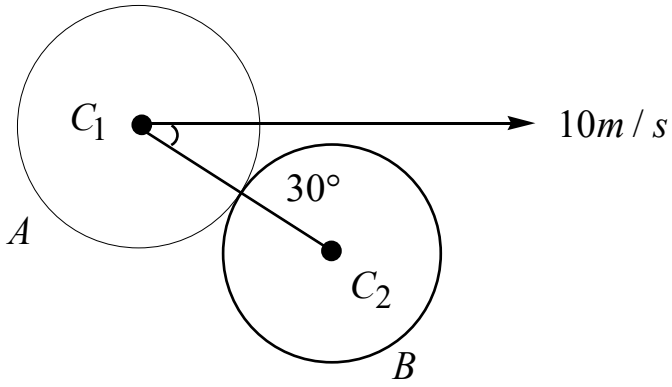
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18. A ball 'A' collides elastically with an another identical ball 'B' at rest with velocity 10m/s at an angle of $30°$ from the line joining their centres C_1 and C_2. Select the correct alternative(s).



A) Speed of ball A after collision in $5m/s$

B) Speed of ball B after collision is $5\sqrt{3}m/s$

C) Both the balls move at right angle to each other after collision

D) Kinetic energy will not be conserved here because collision is not head-on

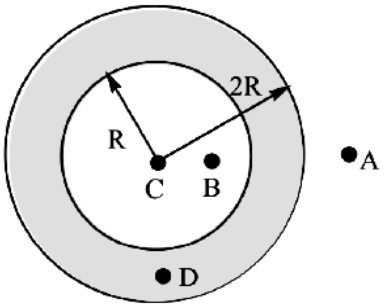
19. The speed of a train increases at a constant rate α from zero to v and then remains constant for an interval and finally decreases to zero at a constant rate β. The total distance travelled by the train is ℓ. The time taken to complete the journey is t. Then,

A) $t = \dfrac{\ell(\alpha+\beta)}{\alpha\beta}$ **B)** $t = \dfrac{\ell}{v} + \dfrac{v}{2}\left(\dfrac{1}{\alpha}+\dfrac{1}{\beta}\right)$

C) t is minimum when $v = \sqrt{\dfrac{2\ell\alpha\beta}{(\alpha+\beta)}}$ **D)** Minimum value of t is $\sqrt{\dfrac{2\ell(\alpha+\beta)}{\alpha\beta}}$

20. Consider a solid neutral conducting sphere of radius 2R having a concentric cavity of radius R. Points A, D, B are at distances 3R, $\dfrac{3R}{2}$ and $\dfrac{R}{2}$ from centre C respectively. S_1 and S_2 are inner surface and outer surface of the hollow sphere.

Mark the **CORRECT** statement(s):



A) If a charge Q is placed at B, potential at point A is $\dfrac{kQ}{3R}$

B) If a charge Q is placed at B, potential at point D is $\dfrac{kQ}{2R}$

C) If charge Q is placed at A, electric field at B and D due to induced charges on the surface is same

D) If charge Q is placed at C and a similar charge Q is placed at A, surface density on the inner surface and outer surface is uniform

SECTION – II
(SINGLE CORRECT ANSWER TYPE)

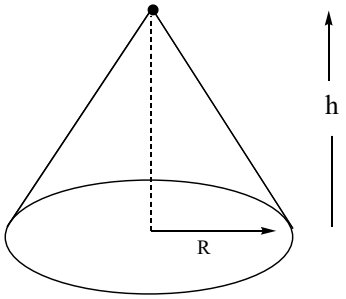
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- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
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Negative Marks : −1 In all other cases

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21. What is moment of inertia of a solid uniform cone of mass m, radius R and height h, about an axis along the diameter of its base



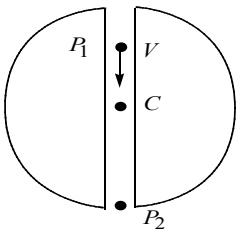
A) $\frac{3}{20}mR^2+\frac{3}{5}mh^2$ **B)** $\frac{3}{20}mR^2+\frac{53}{80}mh^2$

C) $\frac{3}{20}mR^2+\frac{1}{10}mh^2$ **D)** $\frac{1}{2}mR^2$

22. To find the area of a rectangle, it's length is measured to be 12.5 cm and breadth is measured to be 2cm. What is the area of rectangle that should be reported?

A) $25.0cm^2$ **B)** $25cm^2$ **C)** $20cm^2$ **D)** $30cm^2$

23. A tunnel is dug inside the earth of radius R along the diameter as shown. A particle is projected from P_1 at a distance R/2 from the centre of the earth with speed $v=\sqrt{gR}$ towards the centre as shown. Find the time taken by the particle to go to the other end (P_2) of the tunnel



A) $\sqrt{\frac{R}{g}}\left[\sin^{-1}\left(\frac{1}{2}\right)+\sin^{-1}(1)\right]$ **B)** $\sqrt{\frac{R}{g}}\left[\sin^{-1}\left(\frac{1}{\sqrt{5}}\right)+\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)\right]$

C) $\sqrt{\frac{R}{g}}\left[\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)+\sin^{-1}\left(\frac{1}{2\sqrt{5}}\right)\right]$ **D)** $\sqrt{\frac{R}{g}}\left[\sin^{-1}\left(\frac{1}{\sqrt{5}}\right)+\sin^{-1}\left(\frac{3}{2\sqrt{5}}\right)\right]$

24. A 1mH inductor and a $1\mu F$ capacitor are connected in series. The current in the circuit is described by i = 20t, where t is in seconds and *i* is in ampere. The capacitor is initially uncharged. The time when the energy stored in the capacitor first exceeds that in the inductor is

A) $43.2\mu s$ **B)** $53.2\mu s$ **C)** $63.2\mu s$ **D)** $23.2\mu s$

SECTION-III

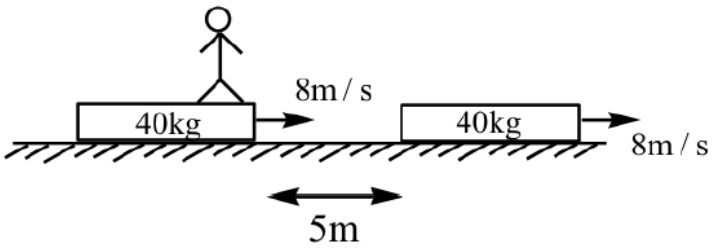
(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases..

25. A boy of mass 40kg is on a plank of mass 40kg moving to right on a smooth floor at 8m/s. He wants to jump on to an identical plank travelling in the front at the same speed. With what minimum speed (in m/s) relative to the plank on which he is standing, should he jump? (Take $g = 10 m/s^2$)



26. A fixed solid cube of mass M, uniformly distributed, is having a very narrow tunnel along one of the body diagonals. Work done in slowly moving a point mass m from centre to the corner is W and the work done in slowly moving the same particle from corner to a very large distance from the cube is W' ,then the ratio of W to W' is given by n. Find n. (Consider every surface is smooth).

27. Suppose electric field in a region is $\overline{E} = kr^3 \hat{r}$ in spherical coordinate system. 'r' is distance from origin and k is a constant. $\hat{r}$ is a unit vector in increasing direction of r. Volume charge density as function of r is given by $\rho = ak\varepsilon_0 r^b$. Find value of a + b.

28. The speed of longitudinal wave is 100 times, then the speed of transverse wave in a brass wire. The Young's modulus of brass is $1.0 \times 10^{11} N/m^2$.

Longitudinal stress in the wire is $x \times 10^7 \frac{N}{m^2}$. Find the value of x.

29. A thin rod of length $\frac{f}{3}$ is placed on the optic axis of a concave mirror of focal length f closest to it, such that its image which is magnified, just touches the rod. The magnitude of magnification of the rod is $\frac{N}{2}$. Find N.

30. A block is kept on a horizontal table. The table is undergoing simple harmonic motion of frequency 3Hz in a horizontal plane. The coefficient of static friction between the block and the table surface is 0.72, Find the maximum amplitude of the table (in cm) for which the block does not slip on surface. (Take $g = \pi^2 m/s^2$)

SECTION – IV
(MATCHING TYPE)

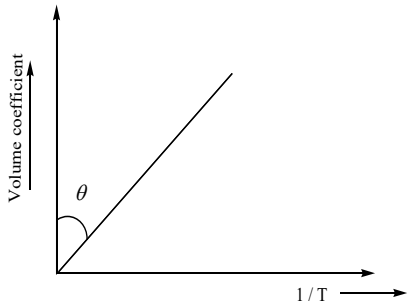
This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
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Negative Marks: –1 In all other cases.

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31. A monatomic ideal gas of 2 moles was found to be undergoing a thermodynamic polytropic process whose volume coefficient is varying with temperature as shown. $\left(R=\frac{25}{3}J/mol.K\right)$. Match the two columns.



List–I		List–II	
P)	Work done by gas in joules, for every 1K rise in temperature, if $\tan\theta=3$	**1)**	3
Q)	Internal energy rise of the gas in joules, for every 1K rise in temperature, if $\tan\theta=2$	**2)**	25
R)	Heat absorbed by the gas in joules, for every 1K rise in temperature, if $\tan\theta=1$	**3)**	$\frac{125}{3}$
S)	Value of θ in degrees for which the internal energy rise of the gas for every 1K rise in temperature is 25J	**4)**	$\frac{50}{9}$

A) P→4; Q→2; R→3; S→1,2,3,4 **B)** P→2,3; Q→1,3,4; R→2; S→3,4

C) P→4; Q→3,4; R→2,3,4; S→1,2 **D)** P→2,4; Q→1,2; R→4; S→1,3,4

32. A particle of mass m = 1kg can move under influence of a conservative force. The potential energy U of the particle under this force varies with its co-ordinate x as $U=-4(x-1)^2(x-2)^2$ where all quantities are in SI units. Column I gives few initial states of the particle, and Column II lists the possibilities regarding its motion. Match them.

Column–I		Column–II	
A)	Particle released at $x=-1m$	**P)**	Will oscillate with an angular

			frequency 2rad/s.
B)	Particle released at $x = +1m$	**Q)**	Will move in negative x direction
C)	Particle kept $x = +1.5m$ and given a very small push in positive x direction	**R)**	Will remain in equilibrium
D)	Particle kept at $x = +1.5m$ and given a very small push in negative x direction	**S)**	Will move to infinity

A) $A \rightarrow P; B \rightarrow Q; C \rightarrow S; D \rightarrow R$ **B)** $A \rightarrow QS; B \rightarrow R; C \rightarrow P; D \rightarrow P$

C) $A \rightarrow PQ; B \rightarrow RS; C \rightarrow P; D \rightarrow R$ **D)** $A \rightarrow Q; B \rightarrow P; C \rightarrow S; D \rightarrow R$

33. A charged particle having non zero velocity is subjected to certain conditions given in List-I, List-II gives possible trajectories of the particle. Match the conditions in List-I with the results in List-II. Given all the fields are non-zero

List–I		**List–II**	
P)	In only uniform electric field	**1)**	The path of the charged particle may be a straight line
Q)	In only uniform magnetic field	**2)**	The path of the charged particle may be a parabola
R)	In uniform magnetic and uniform electric field	**3)**	The path of the charged particle may be circular
S)	Subjected to a net force of constant magnitude	**4)**	The path of the charged particle may be helical with uniform or non-uniform pitch

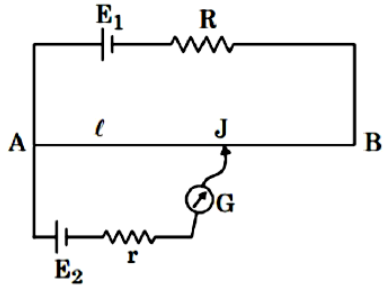
A) $P \rightarrow 1,2; Q \rightarrow 1,3,4; R \rightarrow 1,4; S \rightarrow 1,2,3,4$

B) $P \rightarrow 1,2,3; Q \rightarrow 1,3,4; R \rightarrow 1,4; S \rightarrow 1,2,3$

C) $P \rightarrow 1,2; Q \rightarrow 1,2,3,4; R \rightarrow 1,3,4; S \rightarrow 1,2,3,4$

D) $P \rightarrow 1,2,3; Q \rightarrow 1,3; R \rightarrow 1,3,4; S \rightarrow 1,2,3$

34. In the potentiometer arrangement shown in figure, null point is obtained at length ℓ. Match the following



Column–I		**Column–II**	
P)	If E_1 is increased	**1)**	ℓ should increase
Q)	If R is increased	**2)**	ℓ should decrease
R)	If E_2 is increased	**3)**	ℓ should remain the same
S)	If r is shunted	**4)**	ℓ becomes zero

A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$ **B)** $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 4$

C) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 1; S \rightarrow 3$ **D)** $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$

CHEMISTRY **Max Marks: 60**

SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

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35. Which of the following statement(s) is/are correct regarding the processes involved in the extraction of metals?

A) Roasting of copper pyrite produces blister copper.
B) Calcination of magnesite produces magnesia.
C) In the Mond process, nickel is purified by converting it to nickel carbonyl.
D) Bauxite ore is heated with coke in an electric furnace to produce aluminum.

36. $MeMgCl + H_2C=C=O \longrightarrow A \xrightarrow{H_2O} B \xrightarrow[\Delta]{conc.H_2SO_4} C$

A) Degree of unsaturation in C is 4. **B)** C is an aromatic compound.
C) C is achiral **D)** B is used as solvent.

37. Consider the following reaction sequence:

1) Ethylbenzene is reacted with Br_2 and light $(h\vartheta)$ to form Compound A.
2) Compound A is treated with alcoholic KOH to produce Compound B.
3) Compound B is reacted with O_3 followed by Zn/AcOH to yield Compound C.
4) Compound C is treated with PhMgBr (phenylmagnesium bromide) followed by H_3O^+ to give Compound D.

Which of the following options correctly identifies the major products for Compounds A, B, C, and D respectively?

A) A: *p*-Bromotoluene, B: Benzyl Alcohol, C: Benzoic Acid, D: 1,1-Diphenylethanol
B) A: Benzyl Bromide, B: (o-tolyl)ethane, C: Benzaldehyde, D: Triphenylcarbinol
C) A: 1-Bromo-1-phenylethane, B: Styrene, C: Benzaldehyde, D: Diphenylmethanol
D) A: Benzyl Bromide, B: (o-tolyl)ethane, C: Glyoxal, D: 2-Phenylpropan-2-ol

SECTION – II
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

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38. Green colour Cr^{+3} changes to yellow CrO_4^{-2} on boiling with

A) Acidified H_2O_2 **B)** Acidified Na_2O_2
C) Alkaline H_2O_2 **D)** Acidified $KMnO_4$

39. Which of the following is the most accurate method for determining the degree of dissociation (α) of HF in water?

A) Extrapolating the graph of Λm vs. $\sqrt{c}$ for the electrolyte to zero concentration and using $\Lambda m / \Lambda^{\circ} m$.

B) Determining $\Lambda^{\circ} m$ using Kohlrausch's Law and then using $\Lambda m / \Lambda^{\circ} m$.

C) Measuring the slope of the Λm vs. $\sqrt{c}$ plot for the electrolyte and relating it to the dissociation constant.

D) Directly measuring the slope of the Λm vs. $\sqrt{c}$ plot for the electrolyte and relating it to Ka.

40. Glycine with the structure $\left[\overset{+}{N}H_3 - CH_2 - COO^-\right]$ has $pK_a = 9.8$ and $pK_b = 11.6$.

At what p^H the net charge per glycine molecule is zero?

A) 10.7 **B)** 7.9 **C)** 6.1 **D)** 7

41. $D - glu\cos e \xrightarrow{\text{Re}\, agent} D - gluconic\, acid\,(only)$. This reagent(s) is/are

I) Schiff's reagent II) Fehling solution

III) Bromine in H_2O IV) HNO_3

A) I,II and III only **B)** III only

C) II and III only **D)** I,II,III and IV

SECTION-III

(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

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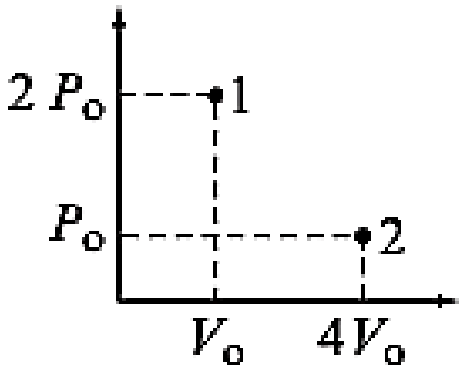
42. A chemist synthesizes a linear silicone polymer by reacting dichlorodimethylsilane $\left[(CH_3)_2 SiCl_2\right]$ with water. They aim to produce polymer chains with an average of 12 silicon atoms each. The polymerization reaction has a 75 % yield. If the chemist wants to actually obtain 6 moles of these average polymer chains, calculate the number of moles of dichlorodimethylsilane they must initially react. Assume only linear polymers are formed.

43. The equation of state for a gas is P(V – nb) = nRT, where b and R are constants. If the pressure and temperature are such that V_m = 10 b, then 1 mole of such gas requires 10 L container. What is the volume (in L) required by 1 mole of the ideal gas at same pressure and temperature?

44. Consider the following reversible reaction A $\rightleftharpoons$ B; $\Delta H = -2$ kcal, the pre-exponential factors are same for the forward and backward reactions. The activation energy of backward reaction is 8 kcal/mol. At 500 K, the fraction of 'A' molecules crossing the

energy barrier for forward reaction is e^{-x}. What is the value of X? Given R = 2 cal $mol^{-1}\ K^{-1}$.

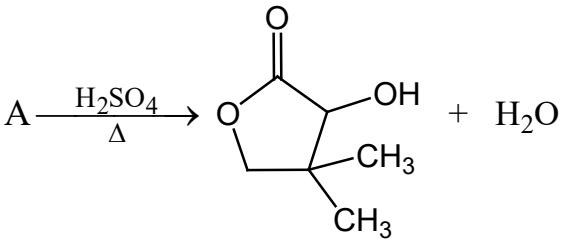
45. A liquid is adiabatically expanded from state – I to state – II, suddenly by a single step, as shown in the figure then



Find the ratio of change in internal energy to change in enthalpy?

46. At 300K, the rate of forward reaction at time t is e^4 times greater than that of reverse reaction for the reaction A + B $\rightleftharpoons$ P. What is the magnitude of free energy change involved at that time (in cal) ? R=2 cal $mol^{-1}K^{-1}$.

47. $C_9H_{17}NO_5 \xrightarrow[H_2O]{OH^-} A + B$



$B \xrightarrow{HNO_2} D + N_2$

$D \xrightarrow{\Delta}$ Acrylic acid

No of secondary carbon atoms in $C_9H_{17}NO_5$ is____

SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases..

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48. Match the reactions (Hydrolyzed product(s)) in List-I with List-II (Compound that undergo hydrolysis) and choose the correct option.

Column–I		Column–II	
A)	H_2 gas evolved	**P)**	CaH_2
B)	Proton donor oxyacid is formed	**Q)**	$POCl_3$
C)	Halogen acid is formed	**R)**	NCl_3
D)	Base is formed	**S)**	B_2H_6
		T)	R_2SiCl_2

A) A → P, B → RT, C → QRT, D → PRT **B)** A → PS, B → QR, C → QT, D → PR

C) A → P, B → RS, C → QRT, D → RT **D)** A → S, B → QST, C → Q, D → PR

49. Match the reactions in List I with nature of the reactions/type of the products in List II.

Column–I		Column–II	
A)	$FeSO_4 \xrightarrow{\Delta}$	**P)**	One of the products is coloured due to charge transfer
B)	$Mn^{2+} + S_2O_8^{2-} + H_2O \rightarrow$	**Q)**	One of the products in +VI oxidation state
C)	$Na_2Cr_2O_7$ (Saturated salt) + H_2SO_4 (conc.) →	**R)**	Redox reaction
D)	$N_2H_4 + CuSO_4 \rightarrow$	**S)**	One of the products is acidic oxide

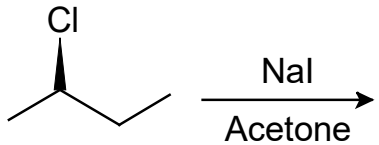
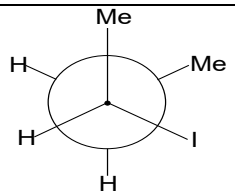
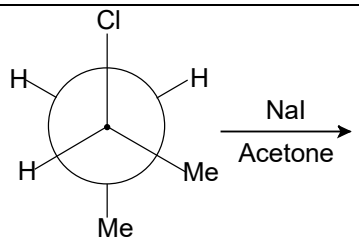
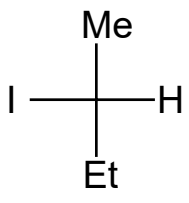
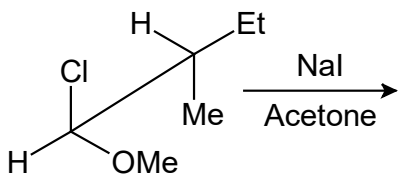
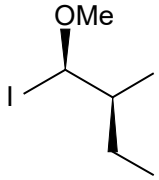
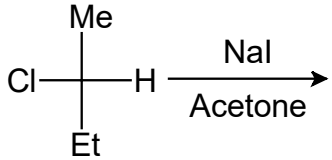
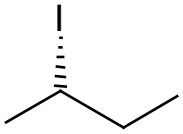
A) A → QS, B → PQ, C → PQ, D → QRS

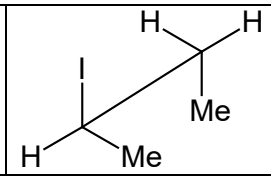
B) A → PRS, B → PQ, C → RS, D → PQR

C) A → QR, B → QS, C → RS, D → PRS

D) A → QRS, B → PQR, C → PQS, D → QR

50. Match the reactions in List I with stereo chemically labelled products in List II.

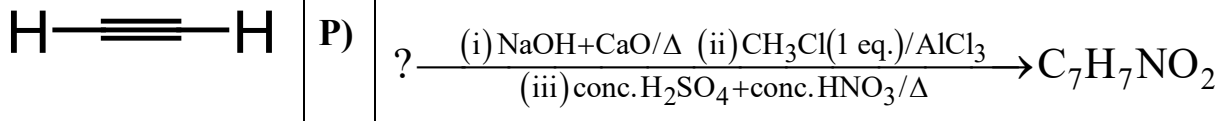
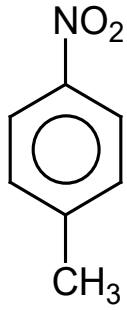
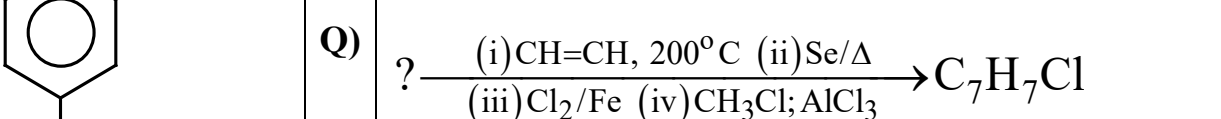
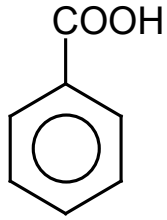
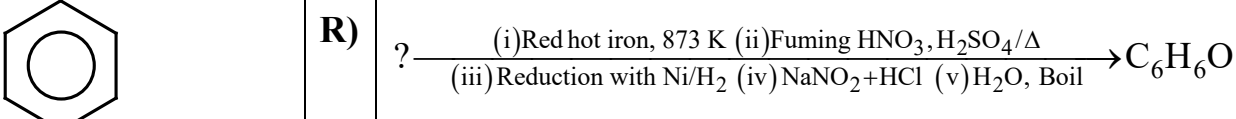
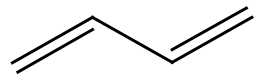
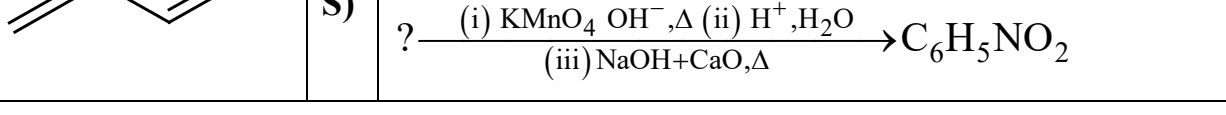
Column–I		Column–II	
A)		**P)**	
B)		**Q)**	
C)		**R)**	
D)		**S)**	

		T)	

A) $A \rightarrow R, B \rightarrow PS, C \rightarrow S, D \rightarrow Q$ **B)** $A \rightarrow S, B \rightarrow P, C \rightarrow R, D \rightarrow Q$

C) $A \rightarrow PS, B \rightarrow RS, C \rightarrow R, D \rightarrow S$ **D)** $A \rightarrow PS, B \rightarrow PS, C \rightarrow R, D \rightarrow S$

51. Match the reactant in List I with sequence of reactions giving suitable product in List II.

Column–I		**Column–II**	
A)		**P)**	Scheme-I 
B)		**Q)**	Scheme-II 
C)		**R)**	Scheme-III 
D)		**S)**	Scheme-IV 

A) $A \rightarrow R, B \rightarrow S, C \rightarrow P, D \rightarrow Q$ **B)** $A \rightarrow R, B \rightarrow P, C \rightarrow S, D \rightarrow Q$

C) $A \rightarrow R, B \rightarrow S, C \rightarrow Q, D \rightarrow P$ **D)** $A \rightarrow S, B \rightarrow P, C \rightarrow Q, D \rightarrow R$